Room Acoustics Summary

Jenna M. Rutowski, Tre DiPassio, Benjamin R. Thompson, Mark Bocko, Michael C. Heilemann

I. Measurements

A calibrated miniDSP UMIK-1 USB microphone was used to measure impulse response of each room [1]. A Mackie SRT215 loudspeaker produced the excitation signal: a 5.5 s long swept sine sampled at 48kHz [2]. Three different loudspeaker positions and three different microphone positions were used to create a total of nine microphone and speaker configurations. This was done to ensure consistency of the recorded data. The impulse response data was recorded by Room EQ Wizard and exported into MATLAB to calculate RT60, C80, and EDT for each room following ISO 3382 standards [3, 4, 5, 6, 7, 8, 9, 10].

II. Room Descriptions

The characteristics of each room can be found in Table I. Room 1 was a (2.4 m × 3.0 m × 2.4 m) rectangular-shaped Whisper Room [11]. Room 2 was a (2.7 m × 3.4 m × 2.4 m) rectangular-shaped mixing room with acoustic treatment. Room 2a did not have windows, while Room 2b did have windows. Room 3 was a (8.8 m × 12.8 m × 2.7 m) rectangular-shaped control room for a recording studio with acoustic treatment and windows. Room 4 was a (3.0 m × 5.5 m × 2.7 m) rectangular-shaped control room for a recording studio with acoustic treatment and windows. Room 5 was a (8.4 m × 15.2 m × 2.7 m) rectangular-shaped classroom with acoustic treatment and no windows. Room 6 was a (7.6 m × 5.5 m × 2.7 m) live room for a recording studio with acoustic treatment. Room 7 was a (7.9 m × 13.4 m × 2.7 m) rectangular-shaped classroom with windows and no acoustic treatment. Room 8 was a (7.8 m × 14.0 m × 7.3 m) live room for a recording studio with sloped ceilings. Room 9 was a (27.4 m × 21.3 m × 12.2 m) hall used for concerts and lectures. Room 10 was a (9.1 m × 13.4 m × 5.5 m) lecture hall with many windows and angled walls. Room 11 was a (111.9 m × 54.9 m × 24.4 m) concert hall. Room 12 was a (28.0 m × 19.8 m × 19.8 m) sanctuary which serves as a chapel. Room 13 was a (15.5 m × 7.8 m × 6.9 m) atrium with many windows and a sloped ceiling. Room 14 was a large rotunda with a radius of 9.1 m and a ceiling height of 24.4 m.

TABLE I
Acoustic Analysis of Rooms

Room	RT60 (s)	C80 (dB)	EDT (s)
1	0.27	20.8	0.18
2a	0.4	20.5	0.15
2b	0.5	20.5	0.20
3	0.5	16.3	0.27
4	0.55	12.3	0.44
5	0.6	9.9	0.54
6	0.65	8.2	0.56
7	0.9	7.5	0.64
8	1.2	5.5	0.86
9	1.5	8.9	0.74
10	1.6	4.7	0.78
11	2	8.9	0.75
12	2.2	6.17	1.37
13	2.2	-0.6	2.01
14	3	-2.0	3.27

References

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